Group 11
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Team Project Proposal: Innovations in Wildfire Management and Prevention

I. INTRODUCTION

Wildfires are uncontrolled fires that rapidly spread across forests and prairies, often ignited by lightning, extreme weather conditions, or human activity. These fires cause devastating damage, destroying natural landscapes, human settlements, and wildlife habitats. In recent years, climate change has intensified wildfire risks by increasing temperatures, extending heat waves, and worsening droughts, all of which create drier fuels that make fires more likely and more severe [1].

One of the most catastrophic wildfires in Canadian history occurred in May 2016 in Fort McMurray, a city located in the heart of the Athabasca oil sands in northeastern Alberta. Surrounded by dense boreal forests, the region was already prone to wildfires, but extreme heat and dry conditions fueled an unprecedented disaster. The fire spread rapidly, doubling in size to 2,656 hectares within hours, leading to a state of emergency and the evacuation of nearly 90,000 residents. By the time the fire was contained on July 5th, it had burned approximately 579,767 hectares of land, destroyed 2,400 homes and businesses, and damaged hundreds of other structures, making it the costliest natural disaster in Canadian history [2].

Fort McMurray remains highly vulnerable to future wildfires due to its dense forest surroundings, increased industrial activity, dry climate, and the increasing impact of climate change. Rising temperatures and prolonged droughts continue to create ideal conditions for fire ignition and rapid spread. Without proactive intervention, Fort McMurray could face similar or even more severe wildfire events in the future [1].

As wildfires become more frequent and intense [1], a proactive approach is essential to reduce their impact. This study proposes using UAV-based cloud seeding [3] to enhance rainfall over high-risk areas before and during the wildfire season, helping to prevent ignition. Combined with AI-driven monitoring, this method offers an adaptive and effective wildfire prevention strategy, improving resilience and response capabilities [4].

II. LITERATURE REVIEW

A. The Impact of Wildfires

Wildfires are one of the most destructive natural disasters in Canada, causing severe environmental, economic, and health impacts. In recent years, climate change has intensified wildfire risks by increasing temperatures, prolonging droughts, and creating drier conditions that fuel more frequent and severe fires [5]. In 2023 alone, wildfires burned approximately 7.8 million hectares of forests across Canada, releasing nearly 3 billion tons of carbon dioxide into the atmosphere [6]. These fires not only destroy vast ecosystems but also force mass evacuations, damage infrastructure, and impose significant economic burdens on affected communities [5]. Additionally, wildfire smoke contains fine particulate matter and toxic gases, leading to respiratory illnesses and worsening air quality far beyond the fire zone [7].

The 2016 Fort McMurray wildfire disaster exemplifies the scale of destruction these fires can cause. The financial cost of the disaster exceeded CAD 9 billion, making it the most expensive natural disaster in Canadian history [1]. As climate change continues to increase wildfire risks, there is an urgent need for more effective prevention and mitigation strategies.

B. Existing Wildfire Prevention and Mitigation Strategies

To address wildfire risks, several prevention and suppression strategies have been implemented. Traditional prevention methods include controlled burns [8], firebreaks [9], and vegetation management [10]. However, these methods are limited by weather conditions, require precise execution, carry risks of unintended fire spread, and demand significant land alteration and ongoing maintenance.

One approach to wildfire prevention is fuel management through grazing. This method can significantly reduce wildfire risks by decreasing fuel biomass. However, it is not universally applicable, as it depends on ecosystem suitability and may lead to overgrazing if not properly managed [11].

Another study discusses various fuel treatment methods, including prescribed burning, mechanical treatment, and biochemical treatment, to mitigate wildfires. Each technique has limitations, such as environmental concerns, high costs, and potential negative impacts on forest ecosystems if not carefully applied [12].

There are more studies that discuss wildfire prevention and mitigation approaches, such as [13-15]. While all of the methods contribute to wildfire prevention and control, their limitations highlight the need for more efficient and adaptable solutions.

C. The Need for a More Effective Approach

Despite these efforts, wildfires continue to increase in frequency and intensity [1], outpacing current prevention and suppression strategies. Traditional methods are often reactive rather than proactive, covering only small areas or struggling to mitigate large-scale fires [16]. Cloud seeding, while promising, is constrained by the need for manned aircraft and favorable weather conditions. Fire detection and suppression methods remain labor-intensive, costly, and limited in their ability to prevent ignition in the first place [3].

Given these challenges, a more adaptive and efficient wildfire management approach is necessary. This study proposes integrating unmanned aerial vehicles (UAVs) with cloud-seeding technology and AI-driven fire monitoring to enhance wildfire prevention and response. By leveraging UAVs, cloud seeding can be conducted more precisely and efficiently, targeting high-risk areas before fire season [3]. At the same time, AI-equipped UAVs can detect early-stage wildfires, map hazard zones, and deploy suppression measures when needed [4]. This dual-function system provides a scalable, proactive solution to mitigate wildfire risks and improve response efforts.

III. PROPOSED SOLUTION

As mentioned before, wildfires often start and spread due to dry conditions, extreme heat, and lightning strikes, causing widespread destruction to forests, wildlife, and communities [1]. Preventing wildfires before they ignite is crucial, and one effective approach is ensuring forests remain moist enough to reduce fire risk. A promising solution is using UAVs for cloud seeding, which can induce rainfall over high-risk areas before peak fire season [3].

Drones equipped with cloud-seeding technology can disperse eco-friendly agents such as sodium iodide [17] or sodium chloride [18] to enhance precipitation, reducing the likelihood of ignition. AI-driven weather monitoring can predict the best seeding windows, ensuring moisture levels remain sufficient when natural rainfall is unlikely. While cloud seeding has been used in water-scarce regions, it has yet to be fully adapted for wildfire prevention. However, challenges exist, including weather dependency [19], environmental concerns about certain seeding materials, and the risk of excessive rainfall leading to floods or landslides [20]. AI optimization can help address these issues by selecting the safest materials, targeting precise areas, and balancing rainfall levels. Additionally, cooperation between regions can ensure fair water distribution while minimizing unintended consequences.

Beyond cloud seeding, UAVs will play a crucial role in wildfire detection and suppression. These drones will be equipped with temperature, humidity, and vegetation sensors, continuously monitoring environmental conditions to assess wildfire risks in real time [4]. By integrating UAVs for both cloud seeding and wildfire management, this comprehensive approach enhances wildfire prevention while ensuring fires are detected and contained quickly if they occur, minimizing environmental, economic, and geopolitical risks.

IV. CONCLUSION

Wildfires are becoming more frequent and severe due to climate change [1], causing widespread destruction to ecosystems, human settlements, and air quality. Given Fort McMurray’s dense forest surroundings, growing industrial activity, dry environment, and the growing effects of climate change, it is still extremely vulnerable to wildfires in the future. Traditional wildfire management strategies struggle to keep pace with these fires' increasing scale and intensity, highlighting the need for more effective solutions [5-7]. This proposal presents a cutting-edge approach to wildfire management by integrating UAV-based cloud seeding [3] and AI-driven wildfire detection and suppression [4]. By combining proactive measures, such as inducing rainfall to prevent fires, with rapid detection and suppression strategies, UAVs offer a scalable and adaptive solution to mitigate the increasing risks posed by wildfires [4]. This dual-function system can enhance preparedness, reduce damage, and support efficient response efforts across vast and difficult-to-access regions. While challenges remain, such as environmental concerns and weather dependency, the innovative use of AI and UAVs provides a promising avenue for improving wildfire resilience and protecting both ecosystems and human settlements from future disasters.

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